SOURAV YADAV

Software Development Engineer | Full-Stack Developer | AI-Integrated Systems

Madhya Pradesh, India | sourav.dev.official@outlook.com

GitHub | LinkedIn | Portfolio | LeetCode | CodeChef

EDUCATION

Indian Institute of Information Technology, Bhopal *Expected 2027*

B.Tech — Electronics & Communication Engineering | CGPA: 7.57/10

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript, TypeScript, Rust, Solidity, SQL

Core CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Backend & Databases: Node.js, Express.js, REST APIs, PostgreSQL, MySQL, Redis, Prisma ORM, JWT Authentication

Frontend Development: React.js, TypeScript, HTML5, CSS3, Tailwind CSS

AI/ML & LLM Engineering: Machine Learning Fundamentals, Deep Learning, Neural Networks, LLM API Integration, Prompt Engineering, Agentic Workflow Design, AI-Assisted Development

Developer Tools: Git, GitHub, Linux, Docker, Vite, Vercel, Unit Testing, Debugging

PROJECTS

FLUX — Intelligent Creative Layout Engine flux-intelligent-creative-layout-en.vercel.app

React · TypeScript · Zustand · Framer Motion

- Engineered a deterministic constraint-satisfaction layout engine that adapts multi-element creatives across device surfaces using priority-based sizing, repositioning, hiding, and AABB collision detection and resolution.
- Decoupled pure-function layout logic (no DOM measurement) from React rendering; stress-tested across 8 distinct device viewports.

SettleOne — Payment Commitment Platform settleone-marketing.settleone.workers.dev

Solidity · TypeScript · React · Node.js · PostgreSQL · Foundry · Viem

- Designed a B2B settlement platform for escrow, delivery verification, dispute resolution, and rule-based settlement, with 10 smart-contract components deployed as upgradeable contracts on 2 EVM testnets (Ethereum Sepolia, Arbitrum Sepolia).
- Structured a 3-application Turborepo monorepo (marketing, transaction workspace, operations dashboard) with shared SDK, API, design-system, type, and utility packages; added AI-assisted risk analysis using blockchain data and configurable policies.

AI Trip Planner — Itinerary Generator ai-trip-planner-v1.vercel.app

React · TypeScript · Vite · Node.js · Express · Google Gemini · Zod

- Implemented an AI itinerary generator that turns free-form descriptions into interactive day-by-day plans, calling Gemini through an Express backend that keeps the API key server-side.
- Validated AI output with Zod; handled malformed JSON, invalid schemas, failed requests, retries, and stale responses; responsive UI.

Protocol Invariant Checker — Rust CLI github.com/Sourav-IIITBPL/protocol-invariant-checker

Rust · CLI · Trait-Based Abstractions · Testing

- Developed a modular Rust CLI that checks protocol invariants against JSON state through a reusable Protocol trait and registry, with state simulation and explanatory output.
- Integrated an ERC-4626 vault module (6 documented invariants), severity-rated findings, and a CI-friendly strict mode; 11/11 tests passing.

PreFlight — Zero-Trust DeFi Transaction Security Platform preflight-v1.vercel.app

Solidity · Foundry · Chainlink CRE · React · TypeScript · Viem

- Architected a hybrid on-chain/off-chain risk engine with 4 modular Solidity guards for 2 DeFi categories (ERC-4626 vaults, Uniswap V2-style AMMs), deployed across 3 testnets (Arbitrum, Ethereum, Base Sepolia).
- Analyzed transactions through fork-based simulation and execution traces to detect suspicious calls, DELEGATECALL activity, and unexpected ERC-20 transfers.

EXPERIENCE

Blockchain Engineering Intern — SKXYWTF *Aug 2026 – Sep 2026, Part-time, Remote, Phoenix, AZ, USA*

- Created an LLM-integrated agentic auditing pipeline that autonomously ran static-analysis tools, interpreted findings through LLM API calls, and generated structured audit reports.

Independent Software Security Researcher (Self-Employed) *July 2025 – June 2026, Remote*

Sherlock · Code4rena · Cantina · Audit Reports

- Audited 20+ smart-contract codebases and identified 11 vulnerabilities (2 High, 3 Medium, 6 Low) by examining business logic, state transitions, edge cases, and security invariants.
- Authored reproducible proof-of-concepts and testing environments to validate each finding through systematic debugging.

ACHIEVEMENTS

- LeetCode Knight — 1892 max rating; 500+ DSA problems solved through consistent practice.
- CodeChef 2-Star Coder — 1543 max rating, built through consistent participation in rated contests.
- HackVision 2026 — Top 100 finalist among 400+ teams.
- Smart India Hackathon 2024 — National-level Participant & Team Lead.
- AlgoUniversity Tech Fellowship — 2024 Stage 1 qualifier; secured Top 4000 among 20,000 participants.
- HER DAO Rust Cohort 1 Graduate.